

CariSurg MedTech Pathways Programme

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1. Structured Triage Workflow Notes

These notes summarise the ED triage process at Mercer General Hospital, based on De Freitas et al. (2020), a qualitative study of patient flow in a Caribbean emergency department in Trinidad and Tobago using the Canadian Triage and Acuity Scale (CTAS).

Key Steps

- Patient arrives at the ED entrance.
- The triage nurse screens at the door and sends Level 1 patients directly to resuscitation.
- Remaining patients proceed to registration with an administrative clerk.
- Vitals are captured: blood pressure, heart rate, SpO2, and temperature, often on paper triage cards.
- The triage nurse reviews the chief complaint and assigns a CTAS level from 1 to 5.
- Basic investigations may be ordered at triage.
- The patient is routed to the appropriate clinical area: Immediate, Urgent, or Minor Care.
- A physician assesses the patient, with timing affected by acuity and department volume.
- Final disposition is admission, discharge, or observation.

Decision Points

- Door screen: Is this a Level 1 patient requiring immediate resuscitation?
- Post-vitals check: Do abnormal vital signs require early escalation?
- CTAS assignment: Which acuity level determines routing and wait time?
- Physician review: Does the clinical picture match the assigned triage level?
- Disposition: Should the patient be admitted, discharged, or held for observation?

Where Information Enters and Gets Lost

Information enters through the patient's verbal complaint, the vital-sign measurements, and the nurse's written triage note. It is most often lost or delayed during paper-to-EHR transcription, during handoff from triage nurse to physician, and when free-text notes are abbreviated under time pressure.

2. First-Pass Workflow Diagram

The diagram shows the ED triage pathway from arrival to disposition and identifies five plausible AI integration points. The core proposed tool sits at the triage nurse assessment stage, immediately after vitals capture and before CTAS assignment.

Workflow Step	Candidate AI Point
Patient arrives	
↓	
Door screen Level 1 patients go to resus	Possible AI point
↓	
Registration Admin clerk records visit	
↓	
Vitals captured BP, HR, SpO2, temperature	Possible AI point
↓	
Triage assessment Chief complaint + CTAS level	Main AI point to test
↓	
Clinical routing Immediate / Urgent / Minor Care	
↓	
Physician assessment	Possible AI point
↓	
Disposition Admit / Observe / Discharge	Possible AI point

(Note that I encountered errors importing the svg into the document so I just created a demo table, wanted to make this work as visually pleasing as possible, please refer to the diagram that would be attached in github, alongside this deliverable)

AI Integration Table

AI Point	Workflow Stage	Function
1	Door screening	Predict queue wait time and pre-flag patients who may deteriorate before assessment.
2	Vitals capture	Detect abnormal combinations of vital signs before formal nurse assessment.
3 (CORE)	Triage nurse assessment	Use NLP on free-text complaints plus a gradient-boosting risk score displayed beside the CTAS level.
4	Physician assessment	Alert if the assigned CTAS level appears inconsistent with vitals and complaint patterns.
5	Disposition	Flag patients statistically likely to deteriorate after discharge based on the visit profile.

3. Candidate Workflow Constraints

Constraint 1: EHR Data Lag. Vitals are captured on paper triage cards before manual entry into the EHR, introducing a 5-30 minute lag. The tool must either interface directly with bedside monitoring or show uncertainty when data may be delayed.

Constraint 2: Nurse Cognitive Load and Alert Fatigue. Triage nurses manage assessments, documentation, and family communication simultaneously. Alert burden must be a primary KPI, and each alert should present the reason and action in under three seconds.

Constraint 3: Free-Text Documentation Variability. Triage notes are rich NLP inputs, but they vary under time pressure. Pre-deployment calibration on local Trinidadian ED text samples is essential.